

STAFFORD, VA, USA

WMO: 720317

Lat: 38.398N

Lon: 77.456W

Elev: 212

StdP: 14.58

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03735

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.5	18.1	-3.1	4.7	20.6	2.0	6.1	24.7	25.2	34.3	22.7	33.9	3.2	320	0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	94.7	76.5	91.2	75.3	89.8	74.3	80.6	87.7	78.7	85.9	77.4	84.8	5.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	151.5	83.7	77.1	142.2	81.6	75.3	133.8	80.0	44.4	88.3	42.4	85.3	40.8	84.4	91.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.7	16.6	13.2	DB	3.8	97.5	5.8	2.5	-0.3	99.3	-3.7	100.8	-6.9	102.2	-11.1	104.0
			WB	3.2	83.1	4.2	2.4	0.2	84.8	-2.2	86.2	-4.5	87.5	-7.6	89.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.8	35.7	38.4	45.8	56.6	65.0	73.1	77.4	75.7	69.2	57.6	46.6	39.7
	DBStd	16.76	10.47	10.27	10.00	9.01	7.92	5.76	4.65	5.14	6.48	8.56	8.29	8.86
	HDD50	1576	459	349	205	32	2	0	0	0	0	24	163	342
	HDD65	4272	908	746	599	281	102	6	0	1	31	260	553	785
	CDD50	4073	16	23	76	232	468	693	848	798	576	261	60	22
	CDD65	1294	0	1	4	31	103	249	383	334	156	32	1	0
	CDH74	12602	0	5	61	391	1048	2461	3824	3154	1375	274	9	0
	CDH80	4984	0	0	6	104	321	949	1713	1339	488	64	0	0
Wind	WSAvg	4.4	4.8	5.5	6.3	5.9	4.5	3.8	3.1	3.1	3.6	4.0	4.4	4.2
Precipitation	PrecAvg	41.2	3.2	2.9	3.8	3.0	3.9	3.3	4.1	3.5	3.8	3.5	3.3	3.3
	PrecMax	59.7	7.6	6.4	9.6	7.0	8.6	10.8	11.7	10.0	10.9	8.0	7.2	7.1
	PrecMin	27.8	0.4	0.3	1.0	0.4	0.6	0.7	0.8	0.5	0.5	0.2	0.2	0.7
	PrecStd	8.1	1.5	1.6	1.9	1.4	1.9	2.0	2.6	2.3	2.5	2.0	1.9	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.8	73.1	80.7	88.5	90.7	96.0	97.6	97.1	93.1	86.6	74.9	70.3
		MCWB	58.5	60.6	62.3	68.8	73.0	77.3	77.9	76.1	75.4	73.2	62.4	62.5
	2%	DB	63.8	67.1	74.8	82.2	87.6	91.3	94.7	93.4	90.1	81.0	70.7	65.1
		MCWB	56.4	57.1	60.3	63.8	72.8	75.2	77.1	76.2	72.9	67.9	60.8	61.8
	5%	DB	58.8	60.8	69.4	78.8	83.7	89.3	91.2	90.4	86.2	76.7	66.6	60.1
		MCWB	52.2	51.4	57.6	62.4	70.0	74.0	76.8	74.4	72.7	67.3	57.9	54.7
	10%	DB	52.5	55.0	63.6	73.8	80.7	86.2	89.6	88.0	81.5	72.6	61.8	54.7
		MCWB	46.6	47.6	53.2	60.7	68.5	73.6	75.5	74.3	69.3	65.4	54.3	49.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.2	64.2	66.3	73.2	78.4	81.3	83.9	82.1	79.3	76.3	66.5	65.7
		MCDB	64.9	69.2	75.5	82.1	84.4	89.0	91.6	88.2	84.9	79.9	71.8	68.1
	2%	WB	57.4	59.1	62.4	69.2	75.4	78.9	81.0	79.4	77.3	73.1	63.1	62.1
		MCDB	63.9	64.7	71.1	76.5	81.6	86.5	88.7	85.7	83.9	76.9	67.6	64.4
	5%	WB	52.5	53.5	59.2	65.6	73.2	77.1	79.1	77.8	75.4	70.2	59.6	56.1
		MCDB	58.2	57.5	66.4	74.6	79.7	84.9	86.9	85.1	81.9	74.8	64.6	58.9
	10%	WB	47.1	48.4	55.2	62.5	71.2	75.4	77.4	76.6	73.2	66.7	55.5	50.3
		MCDB	51.1	54.9	63.2	71.9	77.7	83.1	85.0	84.0	78.2	71.3	60.5	53.4
Mean Daily Temperature Range	5% DB	MDBR	18.7	19.9	21.8	24.4	22.4	22.0	21.1	21.2	21.2	22.0	22.0	18.9
		MCDBR	26.9	27.6	28.8	29.3	26.0	25.6	24.0	24.4	26.3	25.3	27.7	24.1
		MCWBR	20.9	19.1	17.2	14.3	12.7	12.2	10.1	9.9	11.8	14.4	17.9	18.9
	5% WB	MCDBR	23.9	24.2	24.6	24.6	21.9	21.9	21.4	20.5	20.7	20.5	21.4	21.8
		MCWBR	20.5	19.0	17.3	15.1	12.7	11.8	10.7	10.2	10.9	13.5	16.6	19.3
Clear-Sky Solar Irradiance	taub	0.303	0.317	0.352	0.395	0.433	0.488	0.507	0.507	0.417	0.356	0.321	0.307	
	taud	2.507	2.447	2.404	2.304	2.223	2.109	2.079	2.059	2.293	2.463	2.520	2.532	
	Ebn at Noon	280	289	288	280	270	254	248	245	263	271	270	270	
	Edh at Noon	25	30	34	40	45	50	51	51	38	29	24	23	
All-Sky Solar Radiation	RadAvg	686	943	1255	1564	1771	1945	1890	1694	1387	1058	777	605	
	RadStd	61	104	89	115	183	145	97	83	161	119	63	57	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (70 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+243	+148

Nomenclature: See separate page